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Mind, Brain, and Meaning

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CAPTCHARing the Mind:

Inspecting the Distinctiveness of the Human Mind From Computers

The rise of AI, the rapid development of language learning models (LLMs), and the widespread use of machine learning and neural networks in computers have spurred many conversations concerning computers' ability to replace the human brain or assume consciousness. Creations like the Lovelace test assess AI's ability to match human capability, like autonomy or creativity, and AI ethics is largely concerned with the increasing sophistication of AI and the potential for computer behavior to become indistinguishable from humans. To consider whether or not computers can "become" human, we first ask ourselves what do brains and computers share? Can computers do things that brains cannot? Or, what can brains do that computers cannot, and is it possible for computers to perform those functions as technology increases? Although I agree with the statement, "The brain is a computer", I offer the complication that the brain also creates and serves as a vessel for the mind, which computers seem to lack; acknowledging that this extended definition of the brain does not rid the brain of its computer-like qualities, I will use the reduction of the brain to a computer as a starting point to conclude a more arguable statement: the computer is not a mind.

We will first explore why answering the question, "Is the brain a computer?", is rather simple. In lecture, we defined some necessary and sufficient features of a computer: planning/executing, memory storage, processing graphics and sound cards, and managing peripheral appendages. We found these features of a computer easily mappable to parts of the brain that could manage these functions: CPU is to the prefrontal cortex as hard drives and RAM are to the temporal lobe, hippocampus, and prefrontal cortex, etc. While we can map much of the hardware of a computer to a brain, not every component is analogous. But, comparing the material and hardware of a computer and a brain doesn't get us closer to answering our question. When we compare whether or not two pieces of furniture are chairs, do we require that they are built of the same material? No, we focus on what the furniture does for us – do they perform the same functions? Richards & Lillicrap offer a useful definition of the computer: "physical machinery that can implement any computable function in theory" (Richards & Lillicrap, 4). Essentially, computers are physical systems that use programs or algorithms to solve solvable problems. The idea that the problems computers deal with are "solvable" is a caveat we introduced in lecture that tells us that a computer's problems have solutions able to be found by humans, although maybe less efficiently. After

all, humans provide the instructions for the programs run by computers, and computers were built to mimic human cognition. (Lecture 15, Slides 9-15). It is not the solving of a problem that we give the computer to do, but rather the computation of that solution. It seems as though if a computer is modeled off a brain and can only *do* things our human brains know are solvable, the brain is a computer. In other words, the brain, a biological physical system able to not only implement but create computable functions, can do everything a computer can, though with varying speed and efficiency. Richards & Lillicrap echo this idea: “Conceptually, we can perform all of the same operations specified by the languages that we program our laptops with” (Richards & Lillicrap, 4). This thought makes it difficult to disregard the idea that a computer is simply a tool that makes performing mental operations easier. The statement, “the brain is a computer”, then becomes a puzzling one to pose in the first place; we don’t pose the idea “the brain is glasses”, although glasses are a tool that makes visual perception, a cognitive process, easier. To assess the difference between a brain and a computer, I will flip the relationship. To what extent is a computer a brain?

To evaluate whether or not a computer is a brain we now care more about mapping the abilities of the brain to a computer. In addition, the question of what a computer *does* in comparison to a brain becomes more complicated. While computers were built to solve problems, our brains do not just provide us with solutions. Therefore, *how* and *why* certain cognitive processes are performed will be integral to parsing out the potential differences between computers and brains. Throughout this course, we’ve discussed information processing, perception, learning, memory, language, and consciousness as neural processes. Of this non-exhaustive list, I conceive perception, language, and consciousness to be great candidates to explore whether or not an instrumental difference exists between brains and computers. I will begin by attempting to assess whether or not consciousness is a shared feature of both brains and computers. One may be tempted to believe that Patricia Churchland’s idea that consciousness exists in the brain refutes the idea that computers, a non-biological system, have consciousness (Lecture 14, Slide 6). Thomas Nagel’s essay, “What is it Like to Be a Bat”, challenges the idea that we can know what any other form of consciousness looks like: “to deny the reality of logical significance of what we can never describe or understand is the crudest form of cognitive dissonance” (Nagel, 440-441). Nagel says that it is impossible to know what it is like to be a bat because we cannot possibly know what it is like to not live in human bodies with human brains. If a human brain can’t imagine the consciousness of a bat, it seems possible that a human brain would be unable to conceive of a computer consciousness regardless of its existence. Given that I cannot determine whether or not computers and brains share consciousness as an essential constituent, evaluating whether or not a computer is a brain seems out of reach. While validating that, “a computer is a brain” may be too big a project to take on, we can limit our exploration even further

by comparing the mind, a product of the brain, to a computer. Our new statement becomes, “the computer is a mind”.

The aforementioned neural processes I hoped to explore, language and perception, remain viable candidates for exploration; each is a cognitive product of the mind (Lecture 16, Slide 5). Our production and understanding of language is tied to our ability to perceive so we will begin there. Firstly, my discussion of computers for the remainder of this essay will largely be conflated with AI programs and LLMs; computers and the programs that run on computer hardware, like LLMs, will be somewhat interchangeable. Technologies like ChatGPT produce essays and poems that have become increasingly harder to differentiate from human-produced writing. So, if a computer and a human can produce indistinguishable products, do computers produce language the same way our minds do? Bender & Koller introduce a thought experiment in which a hyper-intelligent octopus, O, intercepts a communication signal between two people stranded on islands (Islanders A and B). O begins tapping into the signal and starts to communicate with Islander A after learning much of Islander B’s speech patterns (Bender & Koller, 5188). In lecture, we decided that O would have trouble fooling Islander A. If O can fool Islander A by mimicking human speech, it is only because Islander A believes that O is a member of the “same linguistic community” as them, and therefore Islander A attributes meaning to O’s utterances (Lecture 16, Slide 17). Islander A’s behavior in this case is dictated by communicative intent which Bender & Koller define as something external to language (Bender & Koller, 5187). Replacing O in this thought experiment with a computer, we can see that computers lack communicative intent to inform their language production and understanding. Ultimately, computers do not know what it *means* to produce language nor do they have a motivation for producing it other than being instructed to do so, while human minds do.

In response to Bender & Koller, Mandelkern and Linzen argue that LLMs can refer to the external world with language in a similar way to mind-produced language. They argue that while LLMs’ use of language may not be “grounded” in experience, it can refer to the world and the things it speaks of by being part of a “speech community that uses the word to refer to that thing” (Mandelkern & Linzen, 2). I find this argument convincing, but I don’t believe it to be a refutation of Bender & Koller. After all, when creating and training these models, we are essentially inducting them into our speech community: having them learn words, and read books, articles, and essays the same way we are trained in school to become proficient writers. But, there is still a major difference between our and a computer’s production of language. An LLM being able to reference the thing which it is “speaking” about is not necessarily a matter of understanding, but instead a product of their predictions: “LMs’ words refer too...since they are trained on text whose natural history ties them, eventually, to the referents of the words in question”

(Mandelkern & Linzen, 6). An LLM's ability to reference is reliant on the context in which it learned to predict the word whereas a human mind can speak about Italian mathematician Giuseppe Peano, for example, in a context severed from the instance in which it learned the word in the first place. Examining the example that Mandelkern & Linzen use, Lucy can misattribute information and label mathematician Gödel as Peano, but an LLM would be unable to do this unless it was initially fed information that said Peano proved the incompleteness theorem (Mandelkern & Linzen, 1). Bender and Koller validate this point by explaining LLMs' success in meaning-sensitive tasks: "[LLMs] were instead simply more effective at leveraging artifacts in the data than previous approaches" (Bender & Koller, 1586). So, an LLM's ability to pinpoint a reference for the words it "speaks" does not necessarily mean it can understand language. To clarify, LLMs, even with references, can produce false statements, but for each word produced in that statement, the reference for that word will be true to the LLM's possibly false training data. Again, with the lack of knowledge on computer consciousness, we cannot yet say that LLMs have no personal awareness of true or false. But, we do know that when we ask ChatGPT a question and it gives us wrong information, the model believes that its output is the solution to our question – an LLM is unable to take the data it has been trained on and correct it to reflect truths in the real world. For this correction to happen, there must be facilitated learning and new training data.

On the other hand, this correction and manipulation of data is a process inherent to perception in the human mind. Color constancy is a phenomenon in visual perception that describes this process of the mind well. In Lecture 11 slide 9, we see two [images](#) that, to the human eye, appear to be the same Rubik's cube with differently colored filters overlaid on the top of the image. When we examine the color of each square of the Rubik's cube individually, they are different colors, but our eyes maintain the color we know that square to be under different lighting and color conditions because we have a conceptual understanding of what a Rubik's cube should look like - humans minds can use top-down processing to manipulate sensory data to match the concepts and categorizations we have created about the world and human experience. Similar to the speech communities Mandelkern & Linzen posit exist for language to have meaning, we can see color constancy as a sign of a visual community existing between human minds. Most people with the capacity for visual perception experience color constancy. Similarly, we see a similar effect with object size misperception. The images of concentric circles of differing sizes in Lecture 11 Slide 7 are a common illusion. Most human minds process the inner orange circles in this [image](#) to be different sizes because of the relative sizes of the surrounding shapes. This makes describing the image easier - the left image is a small circle with large circles surrounding it, while the right image is a big circle with tiny circles surrounding it. Without this "misperception", we would have less descriptive explanations- the left image shows large circles with an orange center and the right image shows small circles with an orange center. In this second description, the contrast and discrepancy between the circles'

sizes are understated. Computers do not “misperceive”. Think about the Google CAPTCHAs you are required to complete every so often while browsing the web. You are asked to identify stop lights from somewhat random images of the road or asked to choose which image best represents a child kicking a soccer ball. This data, while initially used to ensure a human was completing the test, has been used to train AI models, enhancing their object recognition. We are required to distinguish stop lights from stop signs, for example, because a computer cannot do this work itself - an AI model cannot create the concept of a stoplight, it can only get better at predicting that an image is or is not a stoplight. Furthermore, if computers are unable to create concepts, then color constancy is not a phenomenon that a computer can experience. A computer’s “concept” of color is the calculation of RGB values present in an image, a bit similar to how our cones detect color. But this is where the mind takes over the brain. Our minds take that sensory data the brain captures, which a computer analogously gets from its RGB calculations, and match it to our preconceptions. We will see a red square on a differently lit Rubik’s cube as red, whereas a computer, unable to have these preconceptions, will see it as orange, purple, or, if trained on hundreds of Rubik’s cube labeled images, a differently colored Rubik’s cube. But in this last case, we must train the computer on more data, maybe through a CAPTCHA, for it to identify the image as a “Rubik’s cube”. Similarly, will see two differently sized orange circles whereas a computer will see two orange circles with different diameters. Our minds do not require extra training or explicit direction to do these things; we are not making predictions based on experiential data, but, instead, our minds unconsciously perceive this way. Since we can’t evaluate computers as brains based on consciousness, the unconscious aspect of misperception, or maybe even perception in general, affords us a unique and necessary quality of the mind that computers do not have.

It’s easy to conflate the surface-level imperfection of AI models today with the imperfection in our perceptual and sensational systems, but the two could not be more different. As we’ve discussed, the imperfection of AI models is entirely consistent with the way LLMs are built and how those models are trained. We see its outputs as imperfect because we compare them with our conception of what the outputs should be. After all, a computer can only solve problems that humans know are solvable. Our imperfections, like misperception, make the world easier to navigate. To not misperceive certain things would arguably place us in a different “visual community” than the one we are already a part of. Ultimately, human brains can *do* everything computers, built and modeled off of the human ability to compute, can. While computers may or may not have a consciousness analogous to a human brain’s consciousness, it is the additional function and qualities of the mind, like perception, that computers are unable to perform and match that establish that a computer is not a mind.

Works Cited

- Bender, Emily M., and Alexander Koller. "Climbing towards NLU: On meaning, form, and understanding in the age of data." *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, <https://doi.org/10.18653/v1/2020.acl-main.463>.
- Mandelkern, Matthew, and Tal Linzen. *Do Language Models Refer?*, 4 Mar. 2024.
- Nagel, Thomas. "What is it like to be a bat?" *The Philosophical Review*, vol. 83, no. 4, Oct. 1974, p. 435, <https://doi.org/10.2307/2183914>.
- Richards, Blake A., and Timothy P. Lillicrap. "The brain-computer metaphor debate is useless: A matter of semantics." *Frontiers in Computer Science*, vol. 4, 8 Feb. 2022, <https://doi.org/10.3389/fcomp.2022.810358>.